

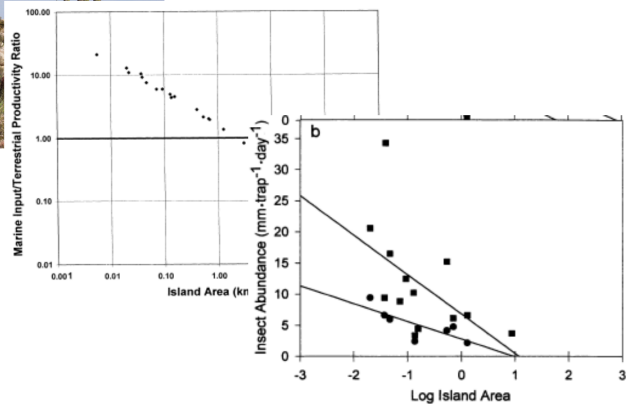
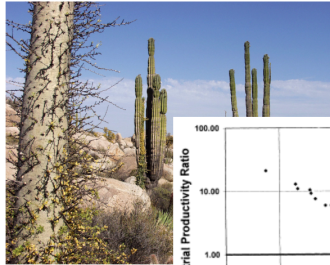


Spatial cascades in metaecosystems

Dominique Gravel

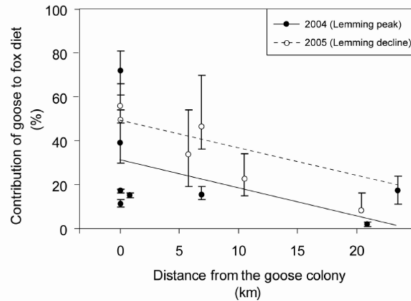


Island secondary productivity





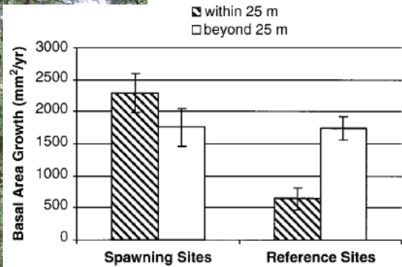
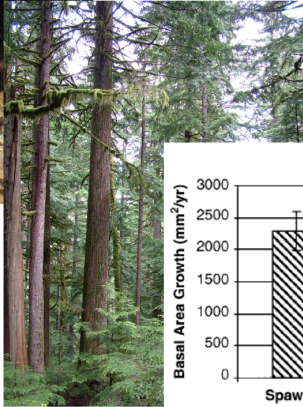
Apparent competition in the arctic



Giroux et al. 2007. JAE.



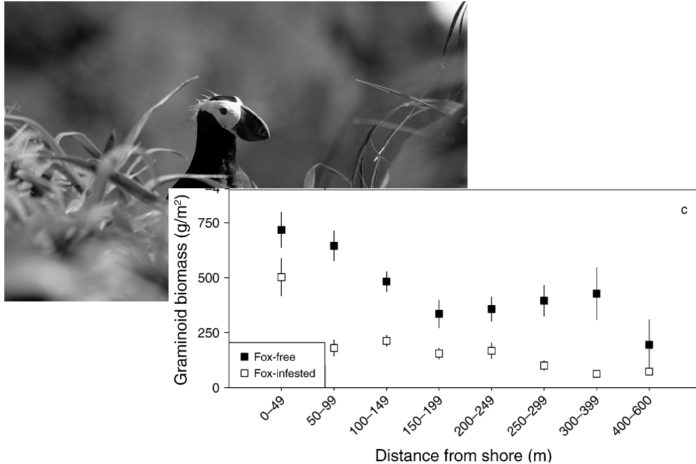
Stream-forest interface



Helfield and Naimoan. 2001.

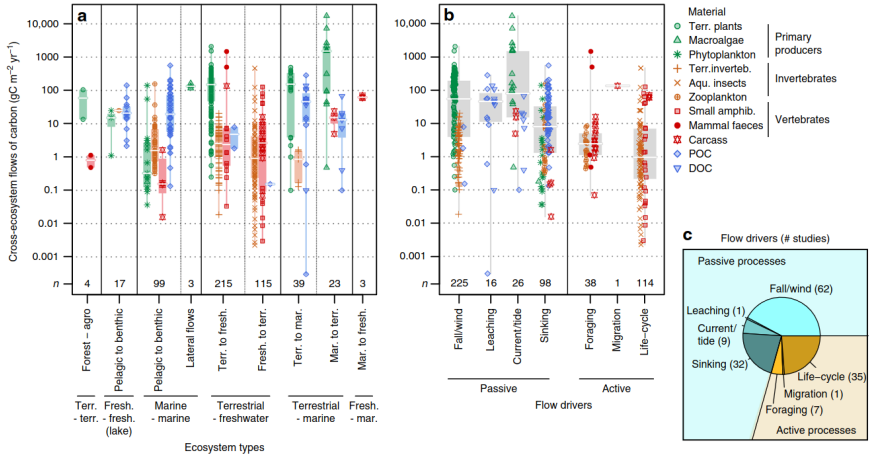


Seabirds colonies



Maron et al. 2006. Ecol. Monogr.

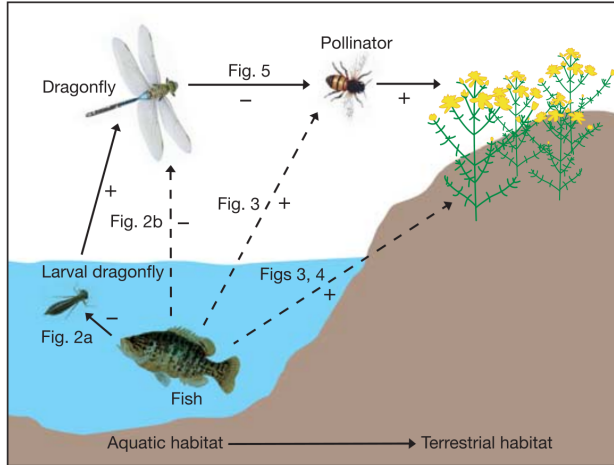
Carbon fluxes Global scale



Gounand et al. 2016. Nat. Communications



Spatial cascades

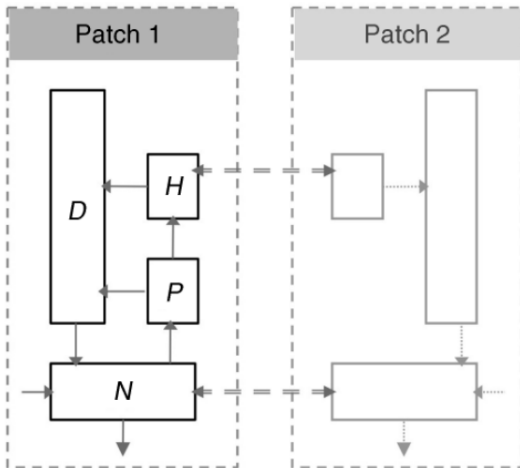


Knight et al. 2005. Nature.

Exercise



A minimal ecosystem model Two patches





A minimal ecosystem model Equations

$$\frac{dN_x}{dt} = I_x + rD_x - e_N N_x - a_P N_x P_x + d_N (N_{x+1} - N_x)$$

$$\frac{dP_x}{dt} = a_P N_x P_x - m_P P_x - e_P P_x - a_H P_x H_x + d_P (P_{x+1} - P_x)$$

$$\frac{dH_x}{dt} = a_H P_x H_x - m_H H_x - e_H H_x + d_H (H_{x+1} - H_x)$$

$$\frac{dD_x}{dt} = m_P P_x + m_H H_x - (r + e_D) D_x + d_D (D_{x+1} - D_x)$$



Exercise

- ▶ Consider a chain of lakes connected by streams
- ▶ Starting with plants and detritus only, look at the distribution of stocks from one location to another
- ▶ Add herbivores, how is it changing ?
- ▶ Decrease progressively the external input I_x , what is happening ?

Source and sink dynamics in metaecosystems



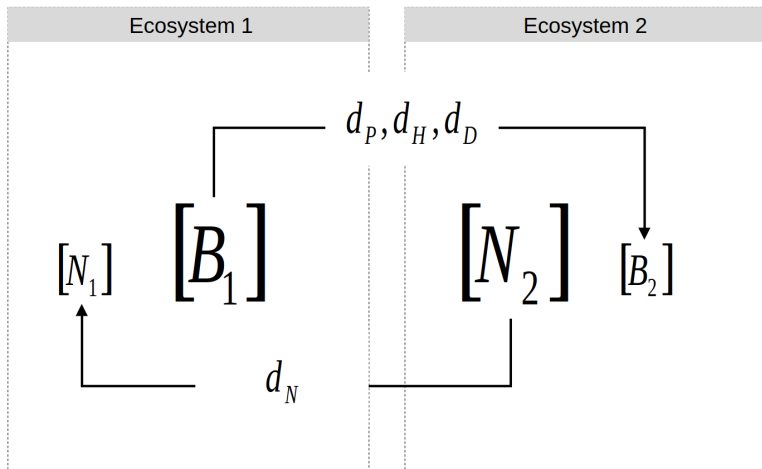
Metaecosystems

Definition : a set of ecosystems connected by spatial flows of energy, materials and organisms across ecosystem boundaries (Loreau et al. 2003, Eco Lett.)

Less appreciated is the fact that these flows may also impose global constraints at the scale of the meta-ecosystem as a whole, thereby generating a strong interdependence among local ecosystems. [] which means that some compartments must be sources whereas others must be sinks in each ecosystem



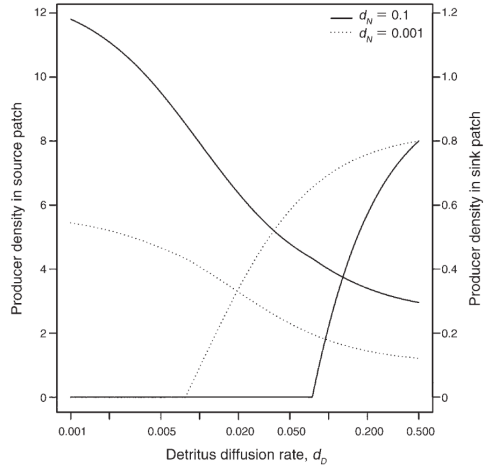
Direction of spatial flows



Gravel et al. 2010. Ecology



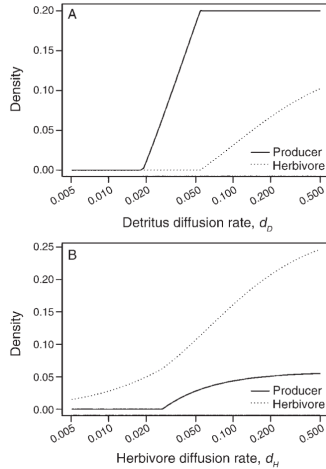
Primary productivity



Gravel et al. 2010. Ecology



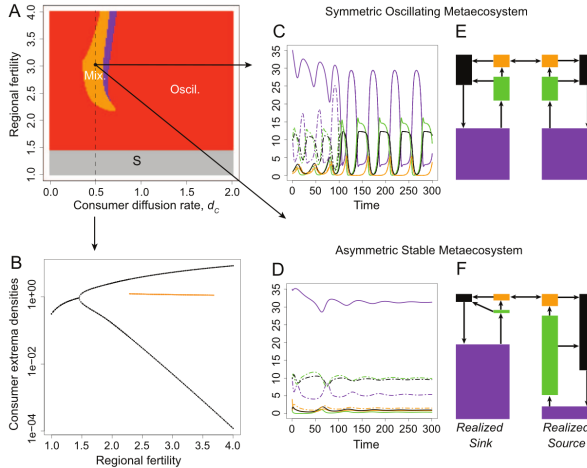
Food chain length



Gravel et al. 2010. Ecology



Paradox of enrichment



Gounand et al. 2014. Am. Nat.



Source-sink dynamics Interpretations

Demographic source/sink : Habitat that is a net exporter/importer of individuals (Pulliam, 1988)

Fundamental source/sink : Habitat in which a species' density-independent growth rate is positive/negative in the absence of spatial flows (Gravel et al. 2010)

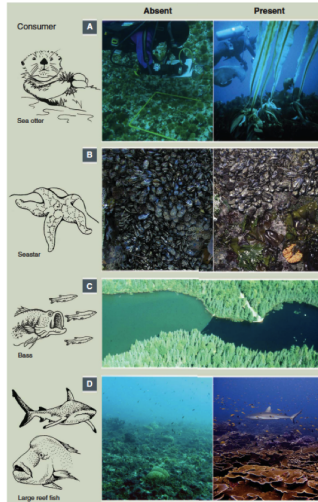
Realized source/sink : Habitat in which a species' density-independent growth rate is positive/negative in the presence of spatial flows (Gravel et al. 2010)

Conditional/Unconditional source/sink : Subsystem that produces/absorbs a surplus of specific living or non-living entities under some/all conditions (Loreau et al. 2013)

Spatial cascades



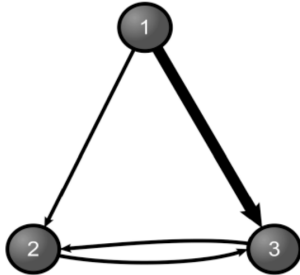
Trophic cascades



Estes et al. 2011. Science



Indirect interactions in food webs



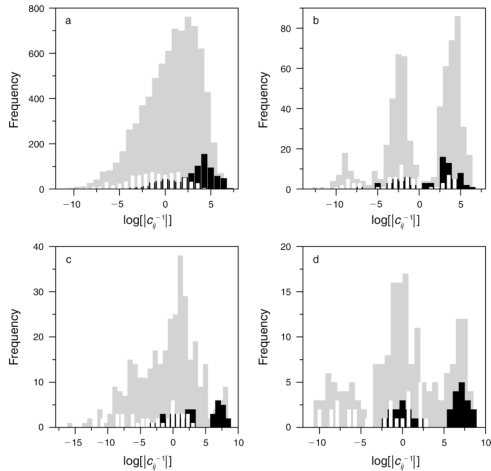
$$\mathbf{C} = \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \\ \begin{bmatrix} 0 & - & - \\ + & 0 & - \\ + & - & 0 \end{bmatrix} \end{array} \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \end{array}$$

$$\mathbf{C}^{-1} = \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \\ \begin{bmatrix} \dots & - & + \\ \dots & \dots & - \\ \dots & \dots & \dots \end{bmatrix} \end{array} \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \end{array}$$

Montoya et al. 2009. Ecology



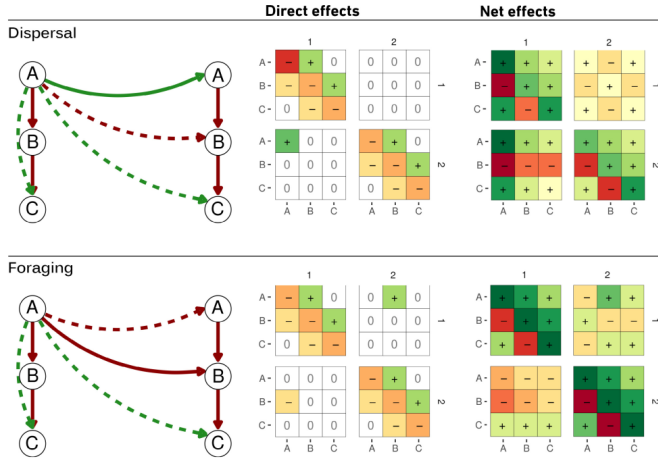
Indirect interactions in food webs



Montoya et al. 2009. Ecology



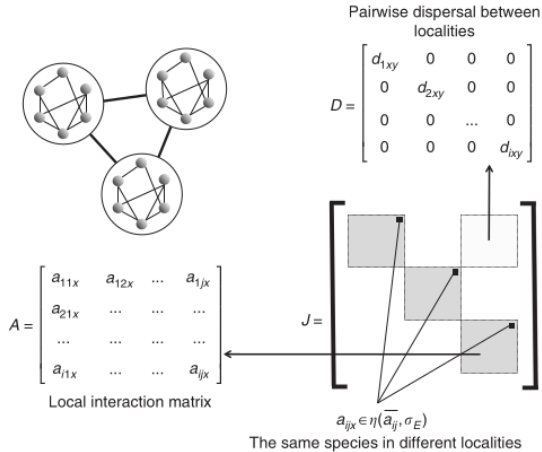
Indirect interactions in food webs Metaecosystems



Garcia et al. 2019. Ecology



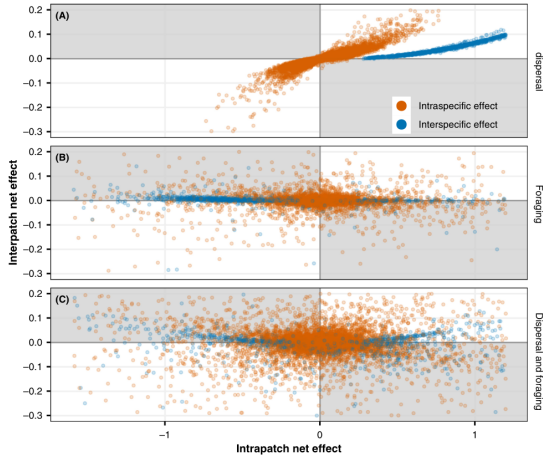
Spatial cascades Model



Gravel et al. 2016. Nat. Communications



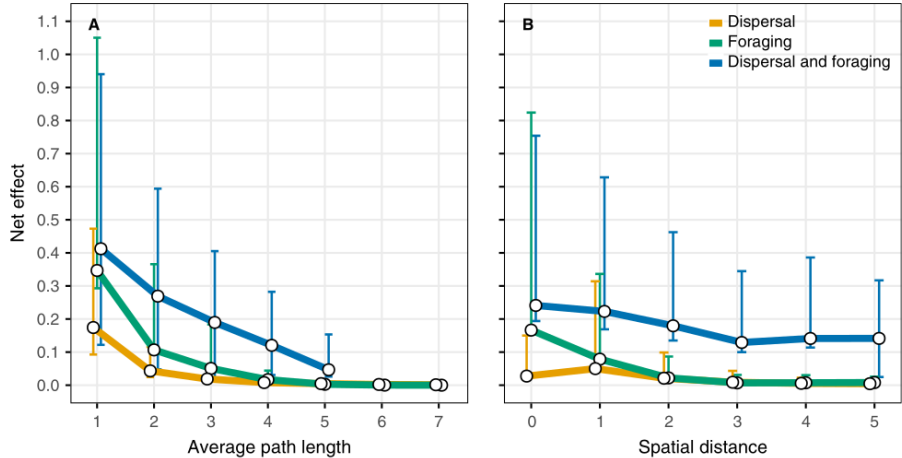
Spatial cascades Emerging indirect interactions



Garcia et al. 2019. Ecology



Spatial cascades Spatial structure



Garcia et al. 2019. Ecology



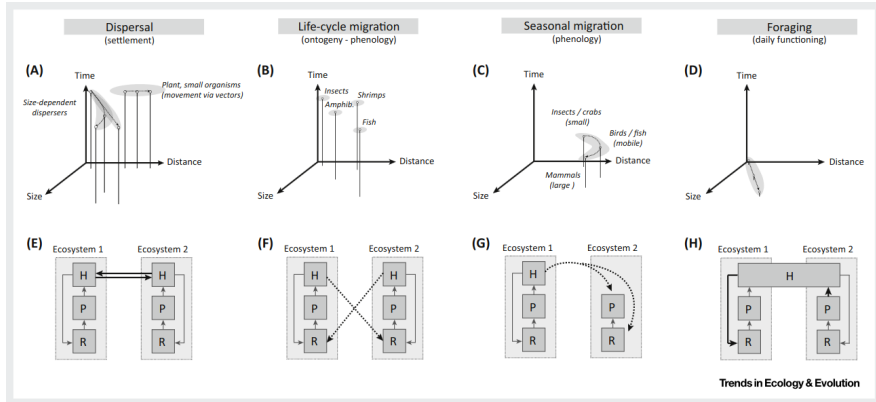
Metaecosystem dynamics Conclusion

- ▶ Local productivity may depend on surrounding ecological context
- ▶ Energy and nutrient flows may alter food web structure and dynamics
- ▶ Contingent source-sink dynamics spatial ecosystems
- ▶ Very conceptual, needs concrete application

Migration as a unique spatial flow



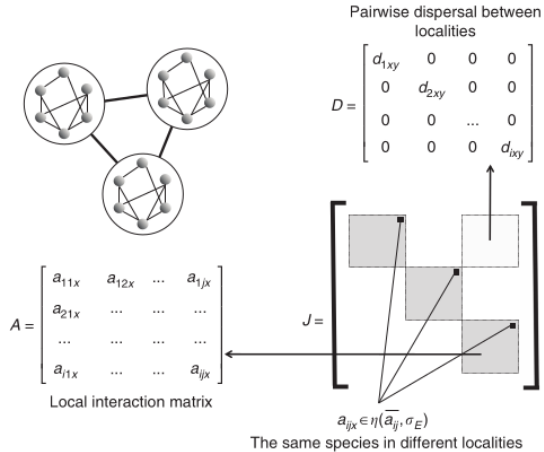
Actual theory



Gounand et al. 2018. TREE



Actual theory



Gravel et al. 2016. Nat. Communications

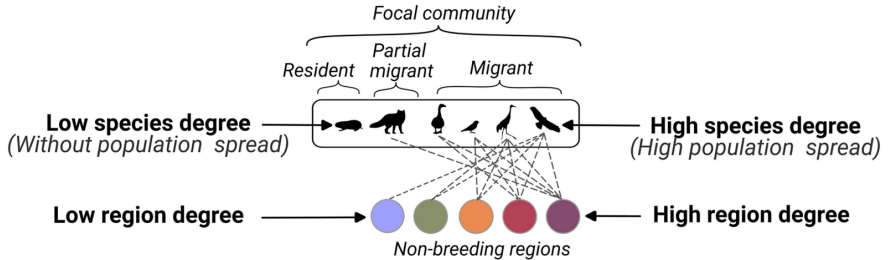


Unique features of migration

- ▶ Happens at very specific moment of the year
- ▶ High frequency and repeated
- ▶ Include all individuals
- ▶ Very short
- ▶ Very species-specific



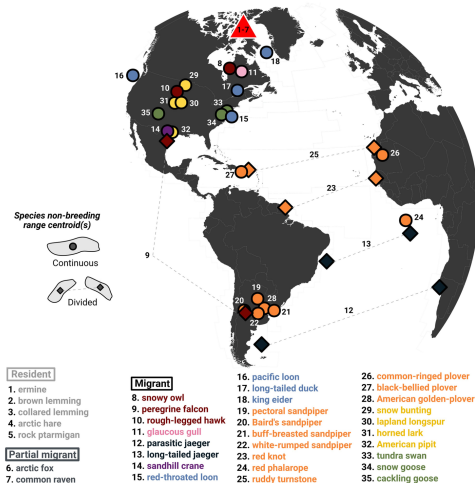
Migration networks



Moisan et al. 2022 *Front. Ecol. Evol.*



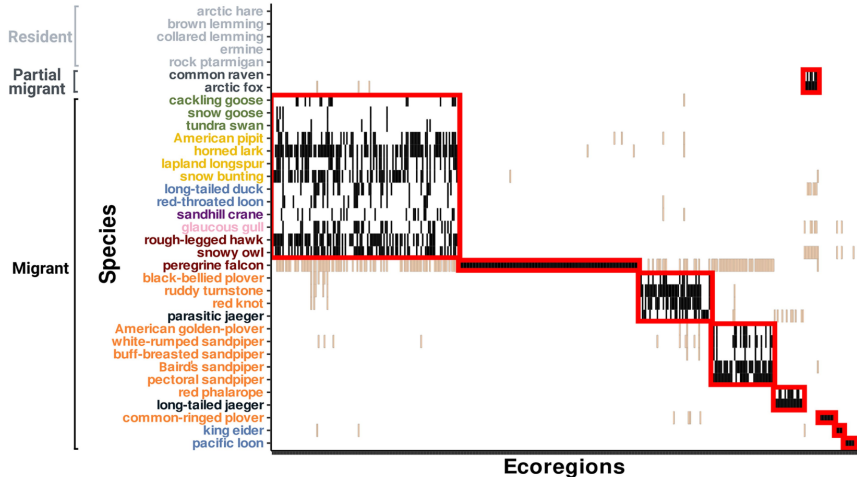
Migration networks



Moisan et al. 2022 *Front. Ecol. Evol.*



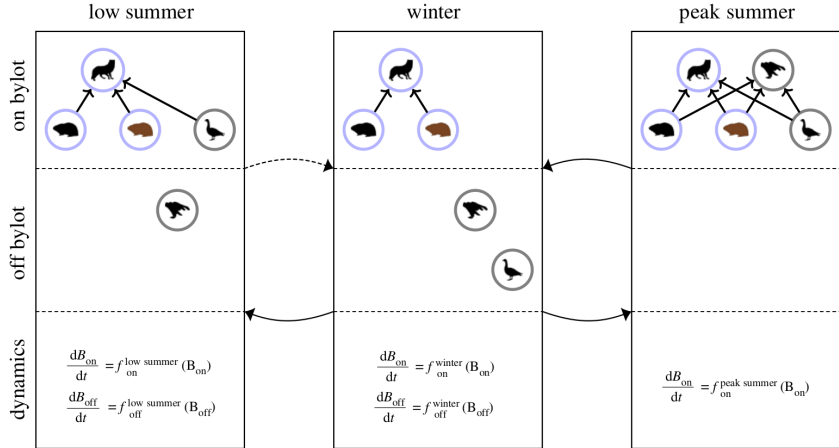
Migration networks



Moisan et al. 2022 *Front. Ecol. Evol.*



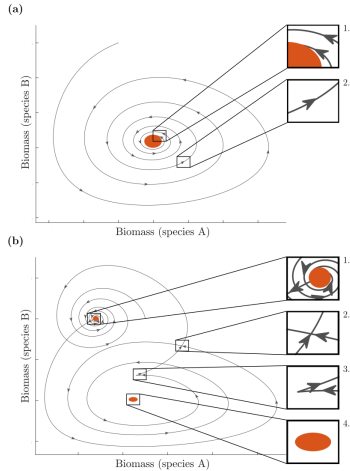
Hybrid models



Hutchison et al. 2020. Phil. Trans. Roy. Soc. A



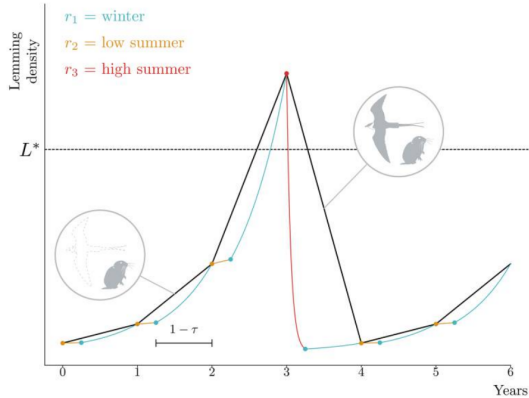
Hybrid models **Stability**



Hutchison et al. 2024. Theoretical Ecology



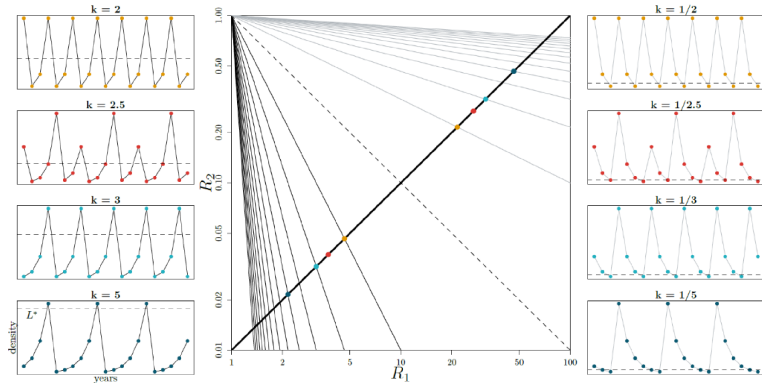
State-dependent switches



Bergeron et al. 2026. Am. Nat



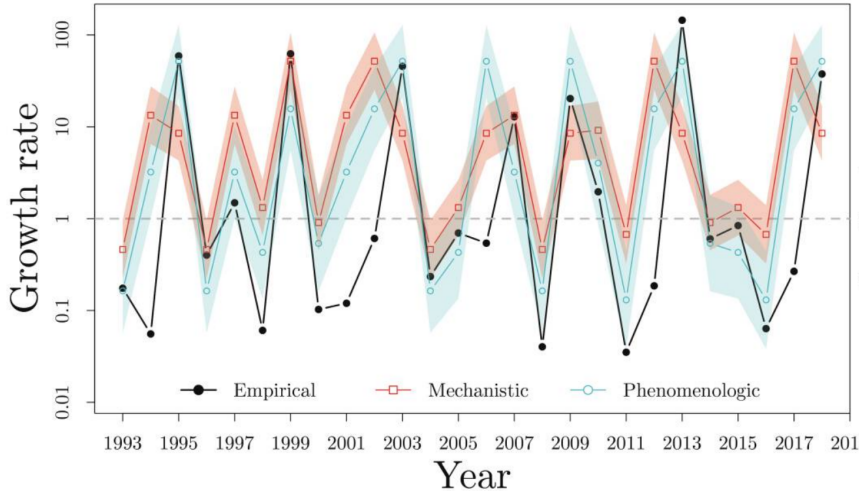
Stability revisited



Bergeron et al. 2026. Am. Nat



Application to lemming cycles



Bergeron et al. 2026. Am. Nat



Open questions

- ▶ What are appropriate metrics to describe and understand such networks ?
- ▶ How disturbances are spreading on such high dimensional networks ?
- ▶ How to track indirect interactions ?